MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY (GENERIC)

END OF SEMESTER EXAMINATION

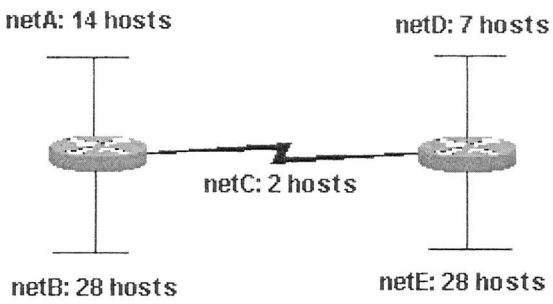
ICT 4703: NEWORK ADMINISTRATION AND INFORMATION SECURITY

TIME ALLOWED: 3 HOURS **DATE: 22nd FEBRUARY 2018**

INSTRUCTIONS

i. This paper contains **TWO** pages. Please check.
ii. There are **SEVEN** questionsin this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

1. Given the Class C network of 204.15.5.0/24, subnet the network in order to create the network in the figure below without address space wastage, with the host requirements shown. **[15 Marks]**



2. Mention any **five** network monitoring tools. **[5 Marks]**
3. Describe the acronym DORA as used in DHCP. **[10 Marks]**

4. ABC Company Limited has a peer to peer network installed with an internet connection from TNM Private Liability company. Over the years they find out that the network configuration is lacking in some crucial areas as far as information management and security is concerned. You are hired to do the advisory work and implement your ideas to come up with a highly reliable and usable network infrastructure. As a network professional with up-to-date knowledge, write the technical proposal to the ABC Company Limited on the new scalable networkand the associated hardware and Software that would solve their current problems and cushion them for the future problems.

[20 Marks]

5. The Directorate of Road Traffic and Safety Services purchased MALTIS software from a South African Company. This software is being used at their five offices across the country. This software is highly dependent on the only VPN connection provided by MTL. You are asked to restructure the network to achieve 99.99% uptime. Briefly present your proposal to management (NOTE: In your proposal include justification on your choice of routing protocols). **[20 Marks]**

6. Describe the RSA key generation Algorithm and use it to encrypt the Message, $M=9$ *given that* $p=5$, $q=11$ *and* $e=3$. **[10 Marks]**

7. You work for tMinds as a Network Administrator, and the organization has three sections, namely: Engineering, Marketing and Sales as shown in the figure 1 below. You would like to segment the already existing cisco-based LAN infrastructure so that

everyone accesses the information of their respective section using Virtual LANs. Assume that you only have one 48-port switch and one router in your LAN:

a. Perform the necessary configuration on the switch and create three VLANs named after the sections. (NOTE: Assign 15 ports to each VLAN (Engineering (1-15), Marketing (16-30) and Sales (31-45)). **[15 Marks]**

b. Describe the importance of VLANs over switched network. **[5 Marks]**

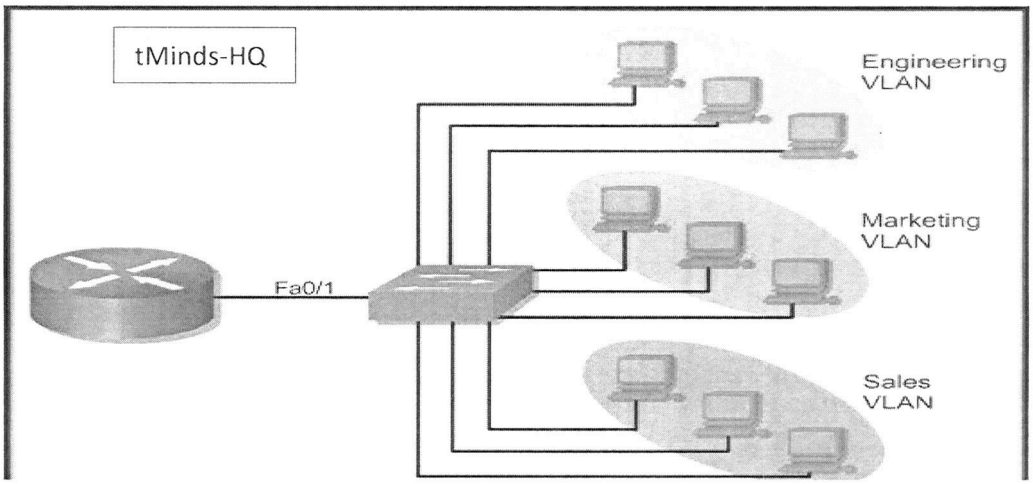


Figure 1: tMinds LAN Infrastructure

END OF QUESTION PAPER